

Obtaining Dibenzalacetone: Aldol Condensation and Purification

Matthew Li* and Cookie Danchaivijitr

Loomis Chaffee High School, CL Organic Chemistry (Section B4)

Instructor: Mr. Osei-Mensah · February 21, 2026

1. Purpose

The purpose of this lab is to utilize the aldol condensation reaction as a way to synthesize new carbon-carbon bonds and obtain a solid final product, dibenzalacetone. In this experiment, we run a double aldol condensation reaction where acetone reacts with benzaldehyde to form dibenzalacetone, using sodium hydroxide as the catalyst. We then isolate and purify the product through recrystallization and calculate percent yields.

2. Data

2.1. Data Table

Table 1: Aldol condensation reaction: materials and measurements.

Measurement	Value
Mass of aldehyde used	2.003 g
Mass of acetone used	0.516 g
Mass of crude dibenzalacetone	1.084 g
Mass of purified dibenzalacetone	[...] g
Melting point of product	[...] °C

3. Calculations

3.1. Crude Percent Yield

$$\begin{aligned}
 n_{\text{acetone}} &= \frac{m_{\text{acetone}}}{\text{M.W.}_{\text{acetone}}} \\
 &= \frac{0.516 \text{ g}}{58.08 \frac{\text{g}}{\text{mol}}} \\
 &= 0.00889 \text{ mol} \\
 n_{\text{benzaldehyde}} &= \frac{m_{\text{benzaldehyde}}}{\text{M.W.}_{\text{benzaldehyde}}} \\
 &= \frac{2.003 \text{ g}}{106.13 \frac{\text{g}}{\text{mol}}} \\
 &= 0.0189 \text{ mol}
 \end{aligned}$$

Limiting reagent: *acetone*

$$\begin{aligned}
 m_{\text{theoretical}} &= n_{\text{limiting}} \times \text{M.W.}_{\text{dibenzalacetone}} \\
 &= 0.00889 \text{ mol} \times 234.30 \frac{\text{g}}{\text{mol}} \\
 &= 2.08 \text{ g}
 \end{aligned}$$

$$\% \text{ Yield}_{\text{crude}} = \frac{m_{\text{crude}}}{m_{\text{theoretical}}} \times 100\% \quad (1)$$

$$\begin{aligned}
 \% \text{ Yield}_{\text{crude}} &= \frac{1.084 \text{ g}}{2.08 \text{ g}} \times 100\% \\
 &= 52.1\%
 \end{aligned}$$

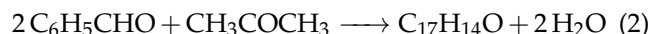
3.2. Pure Dibenzalacetone Percent Yield

$$\begin{aligned}
 \% \text{ Yield}_{\text{pure}} &= \frac{[\dots] \text{ g}}{2.08 \text{ g}} \times 100\% \\
 &= [\dots]\%
 \end{aligned}$$

4. Questions

4.1. Q1. What is the limiting reagent in this aldol reaction? Why?

The limiting reagent in this reaction is acetone. Dibenzalacetone, by its name, contains two "benzaldehyde" pieces attached to one acetone "core". The chemical equation for the reaction is:



Thus, acetone is the limiting reagent.

4.2. Q2. Effect of benzoic acid contamination in old benzaldehyde

If the benzaldehyde is contaminated with benzoic acid, the effect will simply be that we will have less NaOH available to provide a basic enough mixture to provide the necessary condition for the aldol condensation reaction to occur. This is a simple acid-base reaction where the base OH^- will attack the proton on the benzoic acid. Thus, the effective $[\text{OH}^-]$ drops, and there will be less product formed in a slower rate.

5. Evidence of Successful Reaction

1. *The first evidence to support the successful formation of dibenzalacetone is the yellow solid precipitate that formed in the reaction vial during aldol condensation.*
2. *Another evidence is that the purified product appears to be a brighter yellow crystalline solid, with sparkly, well-defined grains on the petri dish.*

6. Aldol Condensation Mechanism Problem

[Add here]

7. Conclusion

In this lab, we performed a mixed aldol condensation reaction between benzaldehyde and acetone to form the product dibenzalacetone using sodium hydroxide as the base. We observed the successful formation of the yellow solid precipitate in the reaction vial during the condensation, and the product was isolated as a yellow precipitated solid, consistent with dibenzalacetone formation [1]. We also obtained a purified, crystalline solid product after recrystallization with ethanol.

Using 0.516g acetone and 2.003g benzaldehyde, acetone was the limiting reagent, giving a theoretical maximum yield of 2.08g dibenzalacetone. We calculated the crude and pure percent yield of the reaction, and found that the values were 52.1% and [...], respectively.

My partners and I have learned, in this lab, to perform aldol condensation, isolation of product, and recrystallization and purity assessment. Further improvements can be made through using freshly distilled (or clearly uncontaminated) benzaldehyde, as old benzaldehyde can be subject to the impact of benzoic acid, which neutralizes some of the NaOH used during the reaction, leading to a potentially lower percent yield.

[Write a conclusion summarizing the lab. Include:

- *A brief summary of the experiment and its main principles (aldol condensation, carbon-carbon bond formation).*
- *Key results: crude and pure percent yields, melting point comparison with the literature value, and overall success of the reaction.*
- *Skills learned during this lab (e.g., recrystallization, vacuum filtration, melting point determination, yield calculations).*

]

References

- (1) Hull, L. A. The Dibenzalacetone Reaction Revisited. *J. Chem. Educ.* **2001**, 78 (2), 226–227, DOI: [10.1021/ed078p226](https://doi.org/10.1021/ed078p226).